Exercise C - II

1. Let z_1, z_2, z_3 are three pair wise distinct complex numbers and t_1, t_2, t_3 are non-negative real numbers such that $t_1 + t_2 + t_3 = 1$. Prove that the complex number $z = t_1z_1 + t_2z_2 + t_3z_3$ lies inside a triangle with vertices z_1, z_2, z_3 or on its boundry.

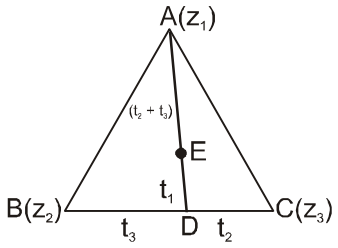
माना z_1, z_2, z_3 तीन भिन्न सम्मिश्र संख्यायें है तथा t_1, t_2, t_3 अऋणात्मक वास्तविक संख्यायें इस प्रकार है कि $t_1 + t_2 + t_3 = 1$ है। सिद्ध कीजिये सम्मिश्र संख्या $z = t_1z_1 + t_2z_2 + t_3z_3$ त्रिभुज z_1, z_2, z_3 के अन्दर अथवा उसकी सीमाओं पर स्थित है।

Ans.

Sol. Affixes of a point D which divides z_2, z_3 in the ratio $t_3 : t_2$ is $\frac{t_2z_2 + t_3z_3}{t_2 + t_3}$ (Internal division)

Affies of a point E which divides. AD in the ratio

$$(t_2 + t_3) : t_1 \text{ is } \left(\frac{t_1z_1 + t_2z_2 + t_3z_3}{t_1 + t_2 + t_3}\right) \quad \text{(Internal division)}$$



Hence E always lies in or on the ΔABC

2. If A, B and C are the angles of a triangle

$$D = \begin{vmatrix} e^{-2iA} & e^{iC} & e^{iB} \\ e^{iC} & e^{-2iB} & e^{iA} \\ e^{iB} & e^{iA} & e^{-2iC} \end{vmatrix} \text{ where } i = \sqrt{-1} \text{ then find the value of D.}$$

यदि A, B तथा C त्रिभुज के कोण है।

$$D = \begin{vmatrix} e^{-2iA} & e^{iC} & e^{iB} \\ e^{iC} & e^{-2iB} & e^{iA} \\ e^{iB} & e^{iA} & e^{-2iC} \end{vmatrix} \text{ जहाँ } i = \sqrt{-1} \text{ तो D का मान ज्ञात कीजिए।}$$

Ans. –4

Sol. $A + B + C = \pi$

$$D = \begin{vmatrix} e^{-i2A} & e^{iC} & e^{iB} \\ e^{iC} & e^{-i2B} & e^{iA} \\ e^{iB} & e^{iA} & e^{-i2C} \end{vmatrix}$$

$= e^{-i2A}\{e^{-i2(B+C)} - e^{i2A}\} - e^{iC}\{e^{-iC} - e^{i(A+B)}\} + e^{iB}\{e^{i(A+C)} - e^{-iB}\}$

$\Rightarrow -(1) + 2e^{i(A+B+C)} - 1$

$\Rightarrow -2 + 2e^{i\pi}$

$\Rightarrow -2-2$

$\Rightarrow -4$

3. If $x = 1+i\sqrt{3}$; $y = 1-i\sqrt{3}$ and $z = 2$, then prove that $x^p + y^p = z^p$ for every prime $p > 3$.

यदि $x = 1+i\sqrt{3}$; $y = 1-i\sqrt{3}$ तथा $z = 2$ तो सिद्ध कीजिए $x^p + y^p = z^p$ है हर एक अभाज्य $p > 3$ के लिए।

Ans.

Sol. $\left(\frac{x}{z}\right)^p + \left(\frac{y}{z}\right)^p = 1$

$(-w^2)^p + (-w)^p = 1$

$0 = 1 + w^p + w^{2p}$

p is prime so not a multiple of 3.

4. Prove that there exists no complex number z such that $|z| < \frac{1}{3}$ and $\sum_{r=1}^{n} a_r z^r = 1$ where $|a_r| < 2$.

सिद्ध कीजिए कि ऐसी कोई सम्मिश्र संख्या नहीं है ताकि $|z| < \frac{1}{3}$ और $\sum_{r=1}^{n} a_r z^r = 1$ जहाँ $|a_r| < 2$ हो।

Ans.

Sol. $1 = |\Sigma a_i z^i| \le \Sigma |a_i z^i|$

$\Rightarrow \quad 1 \le |a_1 z| + |a_2 z^2| + |a_3 z^3| + \ldots\ldots\ldots + |a_n z^n|$

$\Rightarrow \quad 1 < 2\,(|z| + |z|^2 + |z|^3 + \ldots\ldots\ldots + |z|^n)$

$\Rightarrow \quad 1 + |z| + |z|^2 + |z|^3 + \ldots\ldots\ldots\ldots + |z|^n > 3/2$

Case - I $\quad |z| < 1$

$\Rightarrow \quad 1 + |z| + |z|^2 + \ldots\ldots\ldots\ldots \infty > 3/2$

$\Rightarrow \quad \frac{1}{1-|z|} > \frac{3}{2} \quad \Rightarrow \quad 2 > 3 - 3|z|$

$\Rightarrow \quad |z| > 1/3$

Case - II $\quad |z| \ge 1$, then

Obviously $|z| < 1/3$ is not possible.

Hence, $|z| < 1/3$ and $\sum_{i=1}^{n} a_i z^i = 1$ cannot occur simultaneously for any a_i, $|a_i| < 2$.

5. Let $Z = 18 + 26i$ where $Z_0 = x_0 + iy_0$ $(x_0, y_0 \in R)$ is the cube root of Z having least positive argument. Find the value of $x_0 y_0 (x_0 + y_0)$.

यदि $Z = 18 + 26i$ जहाँ $Z_0 = x_0 + iy_0$ $(x_0, y_0 \in R)$, Z का घनमूल है जिसका न्यूनतम धनात्मक कोणांक है तो $x_0y_0(x_0 + y_0)$ का मान ज्ञात करों।

Ans. 12

Sol. $Z = 18 + 26i$

$= 10\sqrt{10}e^{i(\theta)}$ $\qquad \theta = \tan^{-1}\frac{26}{18}$

$Z = \left(10\sqrt{10}\right)^{\frac{1}{3}} e^{i\left(\frac{2K\pi+\theta}{3}\right)} = \tan^{-1}\frac{13}{9}$

$K = 0$

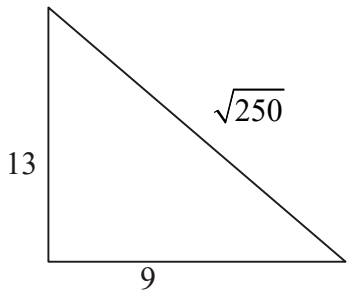
$\Rightarrow Z = \left(10\sqrt{10}\right)^{\frac{1}{3}} e^{i\frac{\theta}{3}}$

$= \left(10\sqrt{10}\right)^{\frac{1}{3}} \left\{\cos\frac{\theta}{3} + i\sin\frac{\theta}{3}\right\}$

$x_0 = \left(10\sqrt{10}\right)^{\frac{1}{3}} \cos\frac{\theta}{3}$

$y_0 = \left(10\sqrt{10}\right)^{\frac{1}{3}} \sin\frac{\theta}{3}$

Now $\tan\theta = \frac{13}{9}$



$\frac{3\tan\frac{\theta}{3} - \tan^3\frac{\theta}{3}}{1 - 3\tan^2\frac{\theta}{3}} = \frac{13}{9}$

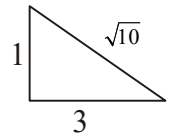
Let $\tan\frac{\theta}{3} = x$

$9(3x - x^3) = 13(1 - 3x^2)$

$27x - 9x^3 = 13 - 39x^2$

$9x^3 - 39x^2 - 27x + 13 = 0$

$\tan\frac{\theta}{3} = \frac{1}{3}$



$\Rightarrow \cos\frac{\theta}{3} = \frac{3}{\sqrt{10}}$

$\sin\frac{\theta}{3} = \frac{1}{\sqrt{10}}$

$x_0 = \left(10\sqrt{10}\right)^{\frac{1}{3}} \cdot \frac{3}{\sqrt{10}}$

$y_0 = \left(10\sqrt{10}\right)^{\frac{1}{3}} \cdot \frac{1}{\sqrt{10}}$

$x_0 y_0 (x_0 + y_0)$

$\left(10\sqrt{10}\right)^{\frac{2}{3}} \cdot \frac{3}{10}\left[\frac{4}{\sqrt{10}}\right]\left(10\sqrt{10}\right)^{\frac{1}{3}}$

$\Rightarrow \frac{10\sqrt{10}}{10\sqrt{10}} = 12$

6. If ω is the fifth root of 2 and $x = \omega + \omega^2$, prove that $x^5 = 10x^2 + 10x + 6$.

यदि ω , 2 का पांचवाँ मूल है तथा $x = \omega + \omega^2$ सिद्ध कीजिये कि $x^5 = 10x^2 + 10x + 6$.

Ans.

Sol. $\omega = 2^{\frac{1}{5}}$

$\omega^5 = 2$

$x^5 = (\omega + \omega^2)^5$

$= \omega^5 (\omega + 1)^5$

$= 2\left[{}^5C_0\omega^5 + {}^5C_1\omega^4 + {}^5C_2\omega^3 + {}^5C_3\omega^2 + {}^5C_4\omega + {}^5C_5\right]$

$= 2\left[2 + 5\omega^4 + 10\omega^3 + 10\omega^2 + 5\omega + 1\right]$

$= 2\left[5\left(\omega^4 + 2\omega^3 + \omega^2\right) + 5\left(\omega^2 + \omega\right) + 3\right]$

$= 2\,[5x^2 + 5x + 3] \qquad = 10x^2 + 10x + 6$

7. Let $z_1, z_2 \in C$ such that $z_1^2 + z_2^2 \in R$. If $z_1(z_1^2 - 3z_2^2) = 10$ and $z_2(3z_1^2 - z_2^2) = 30$, Find the value of $(z_1^2 + z_2^2)$.

यदि $z_1, z_2 \in C$ इस प्रकार है कि $z_1^2 + z_2^2 \in R$. यदि $z_1(z_1^2 - 3z_2^2) = 10$ तथा $z_2(3z_1^2 - z_2^2) = 30$ है, तो $(z_1^2 + z_2^2)$ का मान ज्ञात कीजिए।

Ans. 10

Sol. $z_1, z_2 \in C$

$z_1^2 + z_2^2 \in R$

$z_1^3 - 3z_1z_2^2 = 10 \;\&\; z_2\left(3z_1^2 - z_2^2\right) = 30$

$z_1^3 - 3z_1z_2^2 + iz_2 3z_1^2 + iz_2^3 = 10 + 30i$

$\left[z_1 + iz_2\right]^3 = 10 + 30i$ _____(A)

$\&\ \left[z_1 - iz_2\right]^3 = 10 - 30i$ _____(B)

multiply (A) & (B)

$\left\{\left(z_1 + iz_2\right)\left(z_1 - iz_2\right)\right\}^3 = 1000$

$z_1^2 + z_2^2 = 10$

8. If the equation $(z + 1)^7 + z^7 = 0$ has roots $z_1, z_2, \ldots z_7$, find the value of

(a) $\sum_{r=1}^{7} \mathrm{Re}(Z_r)$ and (b) $\sum_{r=1}^{7} \mathrm{Im}(Z_r)$

यदि समीकरण $(z + 1)^7 + z^7 = 0$ के मूल $z_1, z_2, \ldots z_7$, है तो मान ज्ञात कीजिए।

(a) $\sum_{r=1}^{7} \mathrm{Re}(Z_r)$ तथा (b) $\sum_{r=1}^{7} \mathrm{Im}(Z_r)$

Ans. (a) $-\frac{7}{2}$, (b) zero

Sol. $(z+1)^7 + z^7 = 0$

$\Rightarrow (z+1)^7 = -z^7$

$|z+1|^7 = |-z|^7$

$|z+1| = |z|$

$\Rightarrow$ z is equidistant from (0,0) & (–1,0)

So lies on $x = -\frac{1}{2}$

So all complex roots have real part $-\frac{1}{2}$

So $z_1 + z_2 + z_3 + z_4 + z_5 + z_6 + z_7 = -\frac{7}{2}$ and imaginary parts will be zero.

Thus $\sum_{r=1}^{7} \text{Im}(z_r) = 0$

9. If the expression $z^5 - 32$ can be factorised into linear and quadratic factors over real coefficients as $(z^5 - 32) = (z-2)(z^2 - pz + 4)(z^2 - qz + 4)$ then find the value of $(p^2 + 2p)$.

यदि व्यंजक $z^5 - 32$ को रेखीय तथा द्विघात गुणनखण्डों में (जिसके गुणांक वास्तविक है) में व्यक्त किया गया है $(z^5 - 32) = (z-2)(z^2 - pz + 4)(z^2 - qz + 4)$ तब $(p^2 + 2p)$ का मान ज्ञात करों।

Ans. 4

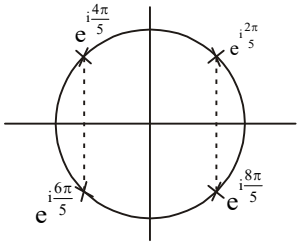
Sol. $z^5 - 32 = 0$

$z^5 = 2^5$

$= 25\ e^{i2k\pi}$

$z = 2e^{\frac{i2k\pi}{5}}$

K = 0,1,2,3,4



$\Rightarrow z^5 - 32 = (z-2)\left(z - 2e^{i\frac{2\pi}{5}}\right)\left(z - 2e^{i\frac{4\pi}{5}}\right)\left(z - 2e^{i\frac{6\pi}{5}}\right)\left(z - 2e^{i\frac{8\pi}{5}}\right)$

$$=(z-2)\left\{z-2e^{i\frac{2\pi}{5}}\right\}\left\{z-2e^{-i\frac{2\pi}{5}}\right\}\left\{z-2e^{i\frac{4\pi}{5}}\right\}\left\{z-2e^{-i\frac{4\pi}{5}}\right\}$$

$$=(z-2)\left\{z^2-2ze^{-i\frac{2\pi}{5}}-2ze^{i\frac{2\pi}{5}}+4\right\}\left\{z^2-2ze^{-i\frac{4\pi}{5}}-2ze^{i\frac{4\pi}{5}}+4\right\}$$

$$p=2\left(e^{-i\frac{2\pi}{5}}+e^{i\frac{2\pi}{5}}\right)$$

$$=2\left\{2\cos\frac{2\pi}{5}\right\}$$

$$\Rightarrow 4\cos\frac{2\pi}{5}$$

$$q=4\cos\frac{4\pi}{5}$$

$$p^2+2p$$

$$16\cos^2\frac{2\pi}{5}+2\cdot4\cos\frac{2\pi}{5}$$

$$16\cos^2\frac{2\pi}{5}+8\cos\frac{2\pi}{5}$$

$$8\left[2\cos^2\frac{2\pi}{5}+\cos\frac{2\pi}{5}\right]$$

$$8\left[2\sin^2 18^\circ+\sin 18^\circ\right]$$

$$8\left[1-\cos 36^\circ+\sin 18^\circ\right]$$

$$8\left[1+\frac{\sqrt{5}-1}{4}-\frac{\sqrt{5}+1}{4}\right]=4$$

10. Let z_i $(i=1,2,3,4)$ represent the vertices of a square all of which lie on the sides of the triangle with vertices (0,0), (2,1) and (3, 0). If z_1 and z_2 are purely real, then area of triangle formed by z_3, z_4 and origin is $\frac{m}{n}$ (where m and n are in their lowest form). Find the value of $(m+n)$.

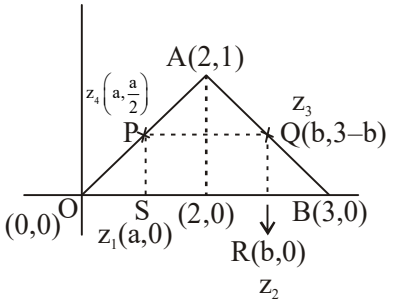
यदि z_i (i = 1, 2, 3, 4) एक वर्ग के शीर्ष को व्यक्त करते है जिनमें से सभी एक त्रिभुज जिसके शीर्ष (0,0), (2,1) तथा (3, 0) की Hkq kvksa ij gS; fn z_1 तथा z_2 पूर्णतया वास्तविक है तब z_3 , z_4 और मूलबिन्दु से बनने वाले त्रिभुज का क्षेत्रफल $\frac{m}{n}$ (जहां m तथा n न्यूनतम समकक्ष रूप में है). (m + n) का मान ज्ञात करों।

Ans. 41

Sol. Equation of line OA

$$y = \frac{1}{2}x$$

$$\therefore \text{ co-ordinates of } P\left[a, \frac{a}{2}\right]$$



Equation of line AB

$$y = \frac{1}{-1}(x-3)$$

Coordinates of Q (b, 3 – b)

Now $\frac{a}{2} = 3 - b$ _____(1)

$b - a = \frac{a}{2}$

$2b = 3a$ _____(2)

Solve (1) and (2)

$\therefore a = \frac{3}{2}, b = \frac{9}{4}$ $\Rightarrow$ side $= \frac{3}{4}$

Ar. of $\Delta PQO = \frac{1}{2}\left(\frac{3}{4}\right)\times\frac{3}{4}$

$= \frac{9}{32}$

$\therefore \frac{9}{32} = m + n$

$m + n = 41$

11. (i) Let C_r's denotes the combinatorial coefficients in the expansion of $(1+x)^n$, $n \in N$. If the integers

$$a_n = C_0 + C_3 + C_6 + C_9 + ...$$
$$b_n = C_1 + C_4 + C_7 + C_{10} + ...$$
and $$c_n = C_2 + C_5 + C_8 + C_{11} + ..., \quad \text{then}$$

prove that (a) $a_n^3 + b_n^3 + c_n^3 - 3a_nb_nc_n = 2^n$, (b) $(a_n - b_n)^2 + (b_n - c_n)^2 + (c_n - a_n)^2 = 2$.

(ii) Prove the identity: $(C_0 - C_2 + C_4 - C_6 + ...)^2 + (C_1 - C_3 + C_5 - C_7 + ...)^2 = 2^n$

(i) यदि Cr's, $(1+x)^n$, $n \in N$ के प्रसार में द्विपद गुणांक है यदि पूर्णांक

$$a_n = C_0 + C_3 + C_6 + C_9 + ...$$
$$b_n = C_1 + C_4 + C_7 + C_{10} + ...$$
तथा $$c_n = C_2 + C_5 + C_8 + C_{11} + ..., \quad \text{तो}$$

fl) d j ks (a) $a_n^3 + b_n^3 + c_n^3 - 3a_nb_nc_n = 2^n$, (b) $(a_n - b_n)^2 + (b_n - c_n)^2 + (c_n - a_n)^2 = 2$.

(ii) सर्वसमिका सिद्ध करो : $(C_0 - C_2 + C_4 - C_6 + ...)^2 + (C_1 - C_3 + C_5 - C_7 + ...)^2 = 2^n$

Ans.

Sol. $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + ... + {}^nC_n x^n$

Let $x = \omega$

$(1+\omega)^n = {}^nC_0 + {}^nC_1 \omega + {}^nC_2 \omega^2 + {}^nC_3 + ... + {}^nC_n \omega^n = (-\omega^2)^n$ _____(1)

$x = \omega^2$

$(1+\omega^2)^n = {}^nC_0 + {}^nC_1 \omega^2 + {}^nC_2 \omega^4 + {}^nC_3 + ... = (-\omega)^n$ _____(2)

Now let $x = 1$

$2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + ... + {}^nC_n$ _____(3)

Add (1), (2) & (3)

$$\left(1+\omega\right)^n + \left(1+\omega^2\right)^n + 2^n = 3\left\{{}^nC_0 + {}^nC_3 + ...\right\}$$

$$\Rightarrow {}^nC_0 + {}^nC_3 + {}^nC_6 + ... = \frac{\left(-\omega^2\right)^n + \left(-\omega\right)^n + 2^n}{3} = a_n$$

Similarly, $\omega^2 \times (1) + \omega \times (2) + (3)$

$$b_n = \frac{2^n + \omega \times \left(-\omega\right)^n + \omega^2 \times \left(-\omega^2\right)^n}{3}$$

$$\text{and } c_n = \frac{2^n + \omega^2 \times \left(-\omega\right)^n + \omega \times \left(-\omega^2\right)^n}{3}$$

We know

$a_n^3 + b_n^3 + c_b^3 - 3a_nb_nc_n = (a_n + b_n + c_n)(a_n + b_n\,\omega + c_n\omega^2)(a_n + b_n\,\omega^2 + c_n\omega)$

$$\left(\frac{3\cdot 2^n}{3}\right)\times\left(\frac{3\left(-\omega^2\right)^n}{3}\right)\times\left(\frac{3\left(-\omega\right)^n}{3}\right)$$

$$=2^n\cdot\left(\omega^3\right)^n=2^n\left(1\right)^n=2^n$$

(b) $a_n^3+b_n^3+c_n^3-3a_nb_nc_n=2^n$

$$\Rightarrow\left(a_n+b_n+c_n\right)\left(a_n^2+b_n^2+c_n^2-a_nb_n-b_nc_n-c_na_n\right)$$

$$\Rightarrow\left(a_n+b_n+c_n\right)\frac{1}{2}\left\{\left(a_n-b_n\right)^2+\left(b_n-c_n\right)^2+\left(c_n-a_n\right)^2\right\}$$

Now $a^n+b^n+c^n=2^n$

Put $x=1$

$2^n=C_0+C_1+C_2+C_3+.....$

$$\therefore 2^n=2^n\cdot\frac{1}{2}\left\{\left(a_n-b_n\right)^2+\left(b_n-c_n\right)^2+\left(c_n-a_n\right)^2\right\}$$

$$\Rightarrow\left(a_n-b_n\right)^2+\left(b_n-c_n\right)^2+\left(c_n-a_n\right)^2=2$$

12. Let z_1, z_2, z_3, z_4 be the vertices A, B, C, D respectively of a square on the Argand diagram taken in anticlockwise direction then prove that :

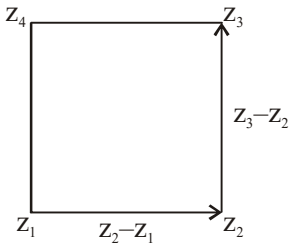
(i) $2z_2=(1+i)z_1+(1-i)z_3$ and (ii) $2z_4=(1-i)z_1+(1+i)z_3$

यदि z_1, z_2, z_3, z_4 सम्मिश्र तल में बने हुये वर्ग के शीर्षो A, B, C, D जो कि वामावर्ती दिशा में लिये गये है, को व्यक्त करते है तो सिद्ध dhft ; s

(i) $2z_2=(1+i)z_1+(1-i)z_3$ तथा (ii) $2z_4=(1-i)z_1+(1+i)z_3$

Ans.

Sol. (i)



$$z_3-z_2=\left(z_2-z_1\right)e^{\frac{i\pi}{2}}$$

$$=(z_2-z_1)i$$

$$z_3-z_2=z_2i-z_1i$$

$$z_2(1+i)=z_3+z_1i$$

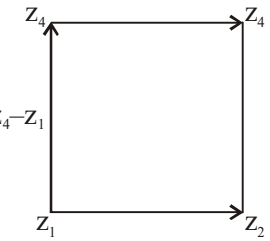
$$z_2 = \frac{z_3}{1+i} + \frac{z_1 i}{1+i}$$

$$z_2 = \frac{z_3(1-i)}{2} + \frac{z_1 i(1+i)}{(1+i)^2}$$

$$= \frac{z_3(1-i)}{2} + \frac{z_1 i(1+i)}{2i}$$

$$2z_2 = z_3(1-i) + z_1(1+i)$$

(ii)



$$z_4 - z_1 = (z_2 - z_1)e^{\frac{i\pi}{2}}$$

$$z_4 = (z_2 - z_1)i + z_1$$

$$z_4 = z_2 i + z_1(1-i) \qquad ...(1)$$

Again

$$z_2 - z_1 = z_3 - z_4$$

$$z_2 = z_3 - z_4 + z_1 \qquad ...(2)$$

from (1) & (2)

$$z_4 = (z_3 - z_4 + z_1)\,i + z_1(1-i)$$

$$z_4(1+i) = z_3 i + z_1(1)$$

$$z_4 = \frac{z_3 i}{1+i} + \frac{z_1}{1+i}$$

$$= \frac{z_3 i(1+i)}{2i} + \frac{z_1(1-i)}{2}$$

$$2z_4 = z_3(1+i) + z_1(1-i)$$

13. Prove that :

(a) $\cos x + {}^nC_1 \cos 2x + {}^nC_2 \cos 3x + ... + {}^nC_n \cos (n+1)x = 2^n \cdot \cos^n \frac{x}{2} \cdot \cos\left(\frac{n+2}{2}\right)x$

(b) $\sin x + {}^nC_1 \sin 2x + {}^nC_2 \sin 3x + ... + {}^nC_n \sin (n+1)x = 2^n \cdot \cos^n \frac{x}{2} \cdot \sin\left(\frac{n+2}{2}\right)x$

सिद्ध कीजिये :

(a) $\cos x + {}^nC_1 \cos 2x + {}^nC_2 \cos 3x + ... + {}^nC_n \cos (n+1)x = 2^n \cdot \cos^n \frac{x}{2} \cdot \cos\left(\frac{n+2}{2}\right)x$

(b) $\sin x + {}^nC_1 \sin 2x + {}^nC_2 \sin 3x + ... + {}^nC_n \sin (n+1)x = 2^n \cdot \cos^n \frac{x}{2} \cdot \sin\left(\frac{n+2}{2}\right)x$

Ans.

Sol. $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + ... + {}^nC_n x^n$

Put $x = e^{i\theta}$

$[1 + e^{i\theta}]^n = {}^nC_0 + {}^nC_1 e^{i\theta} + {}^nC_2 e^{i2\theta} + ... + {}^nC_n e^{in\theta}$

$e^{i\theta}(1 + e^{i\theta})^n = {}^nC_0 e^{i\theta} + {}^nC_1 e^{i2\theta} + {}^nC_2 e^{i3\theta} + ... + {}^nC_n e^{i(n+1)\theta}$

$= \{{}^nC_0 \cos\theta + {}^nC_1 \cos 2\theta + {}^nC_2 \cos 3\theta + ... + {}^nC_n \cos(n+1)\theta\} + 1\{{}^nC_0 \sin\theta + {}^nC_1 \sin 2\theta + ... + {}^nC_n \sin(n+1)\theta\}$

Now $e^{i\theta}\left[1+e^{i\theta}\right]^n$

$e^{i\theta}\left[(1+\cos\theta)+i\sin\theta\right]^n$

$e^{i\theta}\left[2\cos^2\frac{\theta}{2}+i2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right]^n$

$e^{i\theta}2^n\cos^n\frac{\theta}{2}\left\{\cos\frac{\theta}{2}+i\sin\frac{\theta}{2}\right\}^n$

$2^n\cos^n\frac{\theta}{2}\left(e^{\frac{i\theta}{2}}\right)^n \cdot e^{i\theta}$

$2^n\cos^n\frac{\theta}{2}e^{i\theta\left[\frac{n}{2}+1\right]}$

14. Let $f(x) = ax^3 + bx^2 + cx + d$ be a cubic polynomial with real coefficients satisfying $f(i) = 0$ and $f(1 + i) = 5$. Find the value of $a^2 + b^2 + c^2 + d^2$.

माना $f(x) = ax^3 + bx^2 + cx + d$ एक तीन घातीय बहुपद है जिसके गुणांक वास्तविक है, तो $f(i) = 0$ तथा $f(1 + i) = 5$ है। $a^2 + इ^2 + c^2 + d^2$ का मान ज्ञात करें।

Ans. 26

Sol. $f(x) = ax^3 + bx^2 + cx + d \left\langle \begin{matrix} i \\ -i \\ \alpha \end{matrix} \right.$

$ax^3 + bx^2 + cx + d = a[x + i][x - i][x - \alpha]$

$\Rightarrow f(x) = a(x + i)(x - i)(x - \alpha)$

$f(1+i) = a(1+i+i)(1+i-i)(1+i-\alpha) = 5$

$a(1+2i)(1+i-\alpha) = 5$

$a[1+i-\alpha+2i-2-2\alpha i] = 5$

$a[-1+3i-\alpha-2\alpha i] = 5$

$[-a-a\alpha-5]+i[3a-2\alpha a] = 0$

$-a-a\alpha-5 = 0$

$a(1+\alpha) = -5$ ...(1)

$a(3-2\alpha) = 0$

$a \neq 0, \alpha = \frac{3}{2}$

$a = \frac{-5\times 2}{5}$

$a = -2$

$\Rightarrow f(x) = -2(x+i)(x-i)\left(x-\frac{3}{2}\right)$

$= -2\left(x^2+1\right)\left(x-\frac{3}{2}\right)$

$f(x) = -2\left\{x^3 - \frac{3}{2}x^2 + x - \frac{3}{2}\right\}$

$= -2x^2 + 3x^2 - 2x + 3$

$a = -2$ $\qquad\qquad$ $a^2 + b^2 + c^2 + d^2 = 8 + 18 = 26$

$b = 3$

$c = -2$

$d = 3$

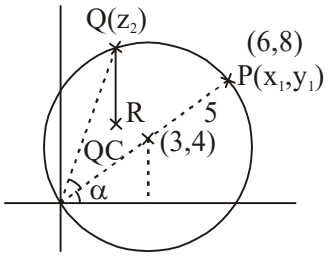
15. A particle starts to travel from a point P on the curve $C_1 : |z-3-4i| = 5$, where $|z|$ is maximum. From P, the particle moves through an angle $\tan^{-1}\frac{1}{3}$ in anticlockwise direction on $|z-3-4i| = 5$ and reaches at point Q. From Q, it comes down parallel to imaginary axis by 2 units and reaches at point R. Find the complex number corresponding to point R in the Argand plane.

एक कण बिन्दु P जो वक्र $C_1 : |z-3-4i| = 5$, पर स्थित है से प्रारम्भ होता है, जहाँ $|z|$ अधिकतम है। P से कण $|z-3-4i| = 5$ पर वामावर्त दिशा में कोण $\tan^{-1}\frac{1}{3}$ से घुमता है तथा बिन्दु Q पर पहुँचता है। Q से काल्पनिक अक्ष के सामानान्तर नीचे की ओर दो इकाई आता है तथा R पर पहुँचता है। सम्मिश्र तल में बिन्दु R के निर्देशांक ज्ञात कीजिये।

Ans. $(3+7i)$

Sol. $|z-(3+4i)| = 5$

$\theta = \tan^{-1}\frac{1}{3}$



$\tan\alpha = \frac{4}{3}$

$\tan(\theta+\alpha) = \frac{\frac{1}{3}+\frac{4}{3}}{1-\frac{4}{9}} = \frac{\frac{5}{3}}{\frac{5}{9}} = 3$

$\frac{y-0}{x-0} = 3$

$\Rightarrow y = 3x$

put in circle

$(x-3)^2 + (y-4)^2 = 5^2$

$(x^2 - 6x + 9) + (9x^2 - 24x + 16) = 25$

$10x^2 - 30x = 0$

$\Rightarrow x = 3, y = 9$

$\Rightarrow z_2 = 3 + 9i$

2 units down

$\Rightarrow 3 + 7i$

16. Evaluate: $\sum_{p=1}^{32} (3p+2)\left(\sum_{q=1}^{10}\left(\sin\frac{2q\pi}{11} - i\cos\frac{2q\pi}{11}\right)\right)^p$.

मान ज्ञात कीजिये : $\sum_{p=1}^{32} (3p+2)\left(\sum_{q=1}^{10}\left(\sin\frac{2q\pi}{11} - i\cos\frac{2q\pi}{11}\right)\right)^p$.

Ans. $48(1 - i)$

Sol. $\sum_{p=1}^{32}(3p+2)\left\{\sum_{q=1}^{10}\sin\frac{2q\pi}{11} - i\cos\frac{2q\pi}{11}\right\}^p$

Now first simplify

$\sum_{q=1}^{10}\sin\frac{2q\pi}{11} - i\cos\frac{2q\pi}{11}$

$\sum -i\left\{-\frac{1}{i}\sin\frac{2q\pi}{11} + \cos\frac{2q\pi}{11}\right\}$

$-i\sum_{q=1}^{10} e^{\frac{i2q\pi}{11}}$

Let $\frac{2\pi}{11} = \theta$

$$-i\left\{\sum_{q=1}^{10} e^{iq\theta}\right\}$$

$$-i\left\{e^{i\theta} + e^{i2\theta} + e^{i3\theta} + ... + e^{i10\theta}\right\}$$

$$-i\left\{\frac{e^{i\theta}\left(1-\left(e^{i\theta}\right)^{10}\right)}{1-e^{i\theta}}\right\}$$

$$-i\left\{\frac{e^{i\theta}\left(1-e^{i10\theta}\right)}{1-e^{i\theta}}\right\} = i$$

Now

$$\sum_{p=1}^{32}(3p+2)i^p$$

$$3\sum_{p=1}^{32}3pi^p + 2\sum_{p=1}^{32}2i^p$$

$3\ \{1i + 2i^2 + 3i^3 + ... + 32i^{32}\} + 2\ \{i + i^2 + ... + i^{32}\}$

$S = 1i + 2i^2 + 3i^3 + ... + 32i^{32}$

$Si = i^2 + 2i^3 + ... + 32i^{33}$

$S(1 - i) = 1i + i^2 + i^3 + ... + i^{32} + (-32)i$

$$S = \frac{-32i}{1-i}$$

$$\Rightarrow \frac{-32i(1+i)}{2}$$

$\Rightarrow -16(i - 1)$

$\Rightarrow 16(1 - i)$

17. Let a, b, c be distinct complex numbers such that $\frac{a}{1-b} = \frac{b}{1-c} = \frac{c}{1-a} = k$. Find the value of k.

माना a, b, c तीन भिन्न सम्मिश्र संख्यायें इस प्रकार है कि $\frac{a}{1-b} = \frac{b}{1-c} = \frac{c}{1-a} = k$ है। k का मान ज्ञात कीजिये।

Ans. $-\omega$ or $-\omega^2$

Sol. $\frac{a}{1-b} = \frac{b}{1-c} = \frac{c}{1-a} = k$

$a = k - bk \qquad b = k - kc \qquad c = k - ak$

$\Rightarrow \quad a = k - (k^2 - k^2c)$

$a = k - (k^2 - (k^3 - ak^3)]$

$a = k - k^2 + k^3 - ak^3$

$a(k^3 + 1) = k[k^2 - k + 1]$

$(ak + a - k)(k^2 - k + 1) = 0$

$k = \frac{-a}{a-1} \qquad$ or $\qquad k^2 - k + 1 = 0$

$k = -\omega, -\omega^2$ and $k = -\frac{a}{a-1}$ is not possible

Hence Proved.

Hindi $\frac{a}{1-b} = \frac{b}{1-c} = \frac{c}{1-a} = k$

$a = k - bk \qquad b = k - kc \qquad c = k - ak$

$\Rightarrow \quad a = k - (k^2 - k^2c)$

$a = k - (k^2 - (k^3 - ak^3)]$

$a = k - k^2 + k^3 - ak^3$

$a(k^3 + 1) = k[k^2 - k + 1]$

$(ak + a - k)(k^2 - k + 1) = 0$

$k = \frac{-a}{1-a} \qquad$ या $\qquad k^2 - k + 1 = 0$

$k = -\omega, -\omega^2$ तथा $k = -\frac{a}{a-1}$ असंभव है।

इति सिद्धम्

18. Let α, β be fixed complex numbers and z is a variable complex number such that, $|z-\alpha|^2 + |z-\beta|^2 = k$. Find out the limits for 'k' such that the locus of z is a circle. Find also the centre and radius of the circle.

माना α, β नियत सम्मिश्र संख्यायें है तथा z एक चर सम्मिश्र संख्या है। $|z-\alpha|^2 + |z-\beta|^2 = k$.

k के सीमान्त मान ज्ञात करें जबकि z का बिन्दुपथ एक वृत है। साथ ही वृत की त्रिज्या एवं केन्द्र भी ज्ञात कीजिये।

Ans. $k > \frac{1}{2}|\alpha - \beta|^2$

Sol. $|z-\alpha|^2 + |z-\beta|^2 = k$

$(z-\alpha)(\bar{z}-\bar{\alpha}) + (z-\beta)(\bar{z}-\bar{\beta}) = k$

$2z\bar{z} - (\bar{\alpha}+\bar{\beta})z - (\alpha+\beta)\bar{z} + |\alpha|^2 + |\beta|^2 - k = 0$

$\Rightarrow z\bar{z} - \left(\frac{\bar{\alpha}+\bar{\beta}}{2}\right)z - \left(\frac{\alpha+\beta}{2}\right)\bar{z} + \frac{|\alpha|^2+|\beta|^2-k}{2} = 0$

center : - $\left(\frac{\alpha+\beta}{2}\right)$ $\qquad \pi = \sqrt{\left|\frac{\alpha+\beta}{2}\right|^2 - \left(\frac{|\alpha|^2+|\beta|^2-k}{2}\right)}$

$$\pi = \sqrt{\frac{2k - \left(|\alpha|^2 + |\beta|^2 - \alpha\bar{\beta} - \beta\bar{\alpha}\right)}{4}}$$

$$\pi = \sqrt{\frac{2k - |\alpha-\beta|^2}{4}}$$

Hence $2k > |\alpha - \beta|2$

$\Rightarrow k > \frac{1}{2}|\alpha-\beta|^2$

19. If the area of the polygon whose vertices are the solutions (in the complex plane) of the equation $x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = 0$, can be expressed in the simplest form as $\frac{a\sqrt{b}+c}{d}$, find the value of $(a + b + c + d)$.

यदि समीकरण $x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = 0$ के हलों को सम्मिश्र तल में निरूपित करने पर बने बहुभुज का क्षेत्रफल सरलतम रूप में $\frac{a\sqrt{b}+c}{d}$ से व्यक्त किया जा सकता है, तो $(a + b + c + d)$ का मान बताइयें।

Ans. 8

Sol. $x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = 0$

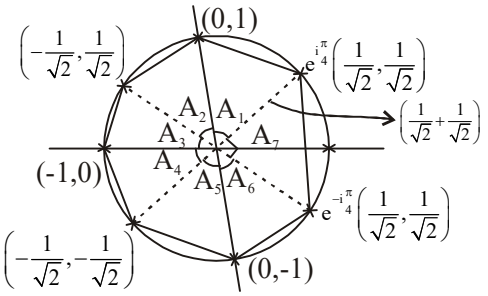
$$\frac{1\left(1-x^8\right)}{1-x} = 0; x \neq 1$$

$1 - x^8 = 0$

$x^8 = 1$

$= \varepsilon^{i2k\pi}$

$x = e^{\frac{i2k\pi}{8}}$



$A_1 + A_2 + A_3 + A_4 + A_5 + A_6 = 6A_1$

$= 6 \cdot \frac{1}{2} \cdot 1 \cdot 1 \cdot \sin\frac{\pi}{4}$

$= \frac{3}{\sqrt{2}}$

$A_7 = \frac{1}{2} \times 1 \times 1 \times \sin\frac{\pi}{2}$

$= \frac{1}{2}$

$\text{Total area} = \frac{3}{\sqrt{2}} + \frac{1}{2}$

$\Rightarrow \frac{3\sqrt{2}+1}{2}$

$a = 3, b = 2, c = 1, d = 2$

$a + b + c + d = 8$

20. If a and b are positive integer such that $N = (a + ib)^3 - 107i$ is a positive integer. Find N.

यदि a तथा b धनात्मक पूर्णांक है इस प्रकार है कि $N = (a + ib)^3 - 107i$ एक धनात्मक पूर्णांक है। N ज्ञात करें।

Ans. 198

Sol. $N = (a+ib)^3 - 107i$

$a^3 - b^3 i + 3abi(a+ib) - 107i$

$a^3 - b^3 i + 3a^2 bi - 3ab^2 - 107i$

$= (a^3 - 3ab^2) + i(3a^2 b - b^3 - 107)$

Now for the integer

$3a^2b - b^3 - 107 = 0$

$3a^2b - b^3 = 107$

$(3a^2 - b^2)\cdot b = 107$

$= 107 \times 1$

$3a^2 - b^2 = 107$ & $b = 1$

$3a^2 = 108$

$a^2 = 36$

$a = 6, b = 1$

$\therefore N = 6^3 - 3\cdot 6\cdot 1^2$

$= 216 - 18$

$= 198$

21. If the biquadratic $x^4 + ax^3 + bx^2 + cx + d = 0$ (a, b, c, d $\in$ R) has 4 non real roots, two with sum $3 + 4i$ and the other two with product $13 + i$. Find the value of 'b'.

यदि एक चार घातीय समीकरण $x^4 + ax^3 + bx^2 + cx + d = 0$ (a, b, c, d $\in$ R) के 4 अवास्तविक मूल है जिनमें दो का योग $3 + 4i$ है तथा अन्य दो का गुणन $13 + i$ है। 'b' का मान ज्ञात करें।

Ans. 51

Sol. Let roots be $\alpha, \beta, \bar{\alpha}, \bar{\beta}$

$\alpha + \beta = 3 + 4i$ $\qquad$ $\bar{\alpha}\bar{\beta} = 13 + i$

$\Rightarrow \bar{\alpha} + \bar{\beta} = 3 - 4i$ $\qquad$ $\Rightarrow \alpha\beta = 13 - i$

$b = \alpha\beta + \alpha\bar{\alpha} + \alpha\bar{\beta} + \bar{\alpha}\beta + \beta\bar{\beta} + \bar{\alpha}\bar{\beta}$

$= \left(\alpha\beta + \bar{\alpha}\bar{\beta}\right) + \left(\alpha\bar{\alpha} + \alpha\bar{\beta} + \bar{\alpha}\beta + \beta\bar{\beta}\right)$

$\left(13 - i + 13 + i\right) + \left(\alpha + \beta\right)\left(\bar{\alpha} + \bar{\beta}\right)$

$26 + (3 + 4i)\,(3 - 4i)$

$26 + 9 + 16 = 51$

22. If ω is the imaginary cube root of unity, then find the number of pairs of integers (a, b) such that $|a\omega + b| = 1$.

यदि ω इकाई का काल्पनिक घनमूल है, तो क्रमागत पूर्णांक युग्मों (a, b) की संख्या बताइयें जिनके लिये $|a\omega + b| = 1$ है।

Ans. 0006

Sol. We have $|a\omega + b| = 1$

$\Rightarrow |a\omega + b|^2 = 1 \Rightarrow (a\omega + b)(a\bar{\omega} + b) = 1 \Rightarrow a^2 + ab(\omega + \bar{\omega}) + b^2 = 1 \Rightarrow a^2 - ab + b^2 = 1$

$\Rightarrow \quad (a-b)^2 + ab = 1 \quad ....(1) \quad (\text{As } 1 + \omega + \omega^2 = 0)$

when $(a-b)^2 = 0$ and $ab = 1$ then $(1, 1); (-1, -1)$

when $(a-b)^2 = 1$ and $ab = 0$ then $(0, 1); (1, 0); (0, -1); (-1, 0)$

Hence **(0, 1); (1, 0); (0, – 1); (– 1, 0); (1, 1); (– 1, – 1) i.e. 6 ordered pairs Ans.**

Paragraph for Question Nos. 23 to 25

Let A, B, C be three sets of complex numbers as defined below.

माना A, B, C सम्मिश्र संख्याओं के तीन समुच्चय है जो निम्न प्रकार से परिभाषित है।

$A = \{z : \text{Im}\, z \geq 1\} \qquad B = \{z : |z - 2 - i| = 3\} \qquad C = \{z : \text{Re}((1-i)z) = \sqrt{2}\}$

23. The number of elements in the set $A \cap B \cap C$ is : **[JEE-2008]**

समुच्चय $A \cap B \cap C$ में सदस्यों की संख्या निम्न है : **[JEE-2008]**

(A) 0 (B) 1 (C) 2 (D) ∞

Ans. B

Sol. $\text{Re}[(1-i)(x+iy)] = \sqrt{2}$

$x + y = \sqrt{2}$

For R

$(x-2)^2 + (y-1)^2 = 9$

$x = 0$

$(y-1)^2 = 5$

$y = 1 \pm \sqrt{5}$

$A \cap B \cap C = \{\text{Point S}\}$

Only $\perp$ point

24. Let z be any point in $A \cap B \cap C$. Then, $|z + 1 - i|^2 + |z - 5 - i|^2$ lies between s : **[JEE-2008]**

(A) 25 and 29 (B) 30 and 34 (C) 35 and 39 (D) 40 and 44

माना z, $A \cap B \cap C$ का कोई एक बिन्दु है। तब $|z+1-i|^2 + |z-5-i|^2$ निम्न के बीच स्थित है : **[JEE-2008]**

(A) 25 तथा 29 (B) 30 तथा 34 (C) 35 तथा 39 (D) 40 तथा 44

Ans. C

Sol. Z = Point S

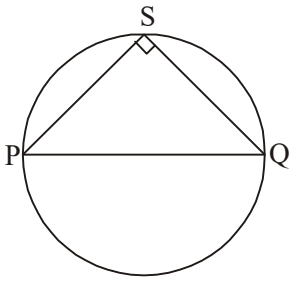
= POI of $x + y = \sqrt{2}$ & $(x-2)^2 + (y-1)^2 = 9$

$|z + 1 - i|^2 + |z - 5 - i|^2 = PS^2 + QS^2$

$= (PQ)^2$

$= 6^2$

$= 36$



25. Let z be any point in $A \cap B \cap C$ and let w be any point satisfying $|w-2-i| < 3$. Then, $|z| - |w| + 3$ lies between: **[JEE-2008]**

(A) – 6 and 3 (B) –3 and 6 (C) – 6 and 6 (D) – 3 and 9

माना z, $A \cap B \cap C$ में कोई एक बिन्दु है तथा w, $|w-2-i| < 3$ को संतुष्ट करने वाला कोई बिन्दु है। तब, $|z| - |w| + 3$ निम्न d scip fLFkr gS **[JEE-2008]**

(A) – 6 तथा 3 (B) –3 तथा 6 (C) – 6 तथा 6 (D) – 3 तथा 9

Ans. BCD

Sol. $|w-2-i| < 3 \Rightarrow$ Points inside circle $B \equiv w$

$|z| - |w| + 3$

$|z-w| < 6 \Rightarrow |z| - |w| \in (-6, 6)$

$|z| - |w| + 3 \in (-3, 9)$

Paragraph for Question Nos. 26 to 28

Let $(1+x)^n = C_0 + C_1x + C_2x^2 + ...+ C_nx^n$. For sum of series $C_0 + C_1 + C_2 + ...$, put x = 1. For sum of series $C_0 + C_2 + C_4 + C_6 + ...$, or $C_1 + C_3 + C_5 + ...$ add or substract equations obtained by putting x = 1 and x = – 1.

For sum of series $C_0 + C_3 + C_6 + ...$ or $C_1 + C_4 + C_7 + ...$ or $C_2 + C_5 + C_8 + ...$ we subsitute x = 1, $x = \omega$, $x = \omega^2$ and add or manupulate results.

Similarly, if suffixes differe by 'p' then we substitute p^{th} roots of unity and add.

माना $(1+x)^n = C_0 + C_1x + C_2x^2 + ...+ C_nx^n$। श्रेणी के योग $C_0 + C_1 + C_2 + ...$ के लिए, x = 1 रखें। श्रेणी के योग $C_0 + C_2 + C_4 + C_6 + ...$, या $C_1 + C_3 + C_5 + ...$ के लिए उन समीकरणों को जोड़े या घटायें जो कि x = 1 तथा x = – 1 रखने पर प्राप्त होती है।

J sh d s; ksx $C_0 + C_3 + C_6 + ...$ या $C_1 + C_4 + C_7 + ...$ या $C_2 + C_5 + C_8 + ...$ के लिए हम x = 1, g ;k ω, $x = \omega^2$ रखते है तथा प्राप्त समीकरणों का योग अथवा कोई अन्य कार्यवाही करते है।

समरूपत:, यदि प्रत्ययों के मध्य 'p' का फर्क हो तो हम इकाई के pवें मूलों को रखते है तथा योग करते है।

26. $C_0 + C_3 + C_6 + C_9 + \ldots\ldots = $

(A) $\frac{1}{3}\left[2^n - 2\cos\frac{n\pi}{3}\right]$ (B) $\frac{1}{3}\left[2^n + 2\cos\frac{n\pi}{3}\right]$

(C) $\frac{1}{3}\left[2^n - 2\sin\frac{n\pi}{3}\right]$ (D) $\frac{1}{3}\left[2^n + 2\sin\frac{n\pi}{3}\right]$

Ans. B

Sol. $(1 + x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 \ldots\ldots$

$2^n = C_0 + C_1 + C_2 + C_3 \ldots\ldots$ .....(1)

$(1 + \omega)^n = C_0 + C_1\omega + C_2\omega^2 + C_3\,\omega^3\ldots\ldots$ .....(2)

$(1 + \omega^2)^n = C_0 + C_1\omega^2 + C_2\omega^4 + C_3\,\omega^6\ldots\ldots$ .....(3)

वदं कक्प्रदह ; 1द्ध ; 2द्ध ; 3द्धजोड़ने पर

$2^n + (1 + \omega)^n + (1 + \omega^2)^n = 3[C_0 + C_3 + C_6 \ldots\ldots]$

$$\frac{2^n + (-\omega^2)^n + (-\omega)^n}{3} = C_0 + C_3 + C_6\ldots\ldots$$

$$C_0 + C_3 + C_6 \ldots\ldots = \frac{2^n + e^{i\frac{n\pi}{3}} + e^{-i\frac{n\pi}{3}}}{3}$$

$$= \frac{1}{3}\left[2^n + 2\cos\frac{n\pi}{3}\right] \qquad \text{(B)}$$

27. $C_1 + C_5 + C_9 + \ldots. = $

(A) $\frac{1}{4}\left[2^n - 2^{n/2}2\cos\frac{n\pi}{4}\right]$ (B) $\frac{1}{4}\left[2^n + 2^{n/2}2\cos\frac{n\pi}{4}\right]$

(C) $\frac{1}{4}\left[2^n - 2^{n/2}2\sin\frac{n\pi}{4}\right]$ (D) $\frac{1}{4}\left[2^n + 2^{n/2}2\sin\frac{n\pi}{4}\right]$

Ans. D

Sol. $(1 + x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + C_4x^4 + C_5x^5 + C_6x^6 \ldots\ldots$

चनजगा ञ 1 रखने पर $2^n = C_0 + C_1 + C_2 + C_3 + C_4 + C_5 + C_6 + C_7 + C_8$ ..... (1)

चनजगा ञ 1 रखने पर $0 = C_0 - C_1 + C_2 - C_3 + C_4 - C_5 + C_6 - C_7 + C_8$ ..... (2)

चनजगा ञ परखने पर $(1 + i)^n = C_0 + iC_1 - C_2 - iC_3 + C_4 + iC_5 - C_6 - iC_7 + C_8$ ..... (3)

चनजग ञ परखने पर $(1 - i)^n = C_0 - iC_1 - C_2 + iC_3 + C_4 - iC_5 - C_6 + iC_7 + C_8$ ..... (4)

Now अब (1) – (2) – i(3) + i(4)

$4(C_1 + C_5 + C_9\ldots\ldots) = 2^n - 0 - i(1 + i)^n + i(1 - i)^n$

$= 2^n + (-i)\,[(1 + i)^n - (1 - i)^n]$

$= 2^n + (-i)\,[(1 + i)^n - (1 - i)^n]$

$$= 2^n - i2^{n/2}\left[e^{i\frac{n\pi}{4}} - e^{-i\frac{n\pi}{4}}\right]$$

$$= 2^n - i\,2^{n/2}\,.\,2i\sin\left(\frac{n\pi}{4}\right)$$

$$\therefore \qquad C_1 + C_5 + C_9 \ldots\ldots = \frac{1}{4}\left[2^n + 2^{n/2}\,.2\sin\frac{n\pi}{4}\right]$$

28. $C_2 + C_6 + C_{10} + \ldots\ldots =$

(A) $\frac{1}{4}\left[2^n - 2^{n/2}\,2.\cos\frac{n\pi}{4}\right]$ (B) $\frac{1}{4}\left[2^n + 2^{n/2}\,2.\cos\frac{n\pi}{4}\right]$

(C) $\frac{1}{4}\left[2^n - 2^{n/2}\,2.\sin\frac{n\pi}{4}\right]$ (D) $\frac{1}{4}\left[2^n + 2^{n/2}\,2.\sin\frac{n\pi}{4}\right]$

Ans. A

Sol. $(1 + x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + C_4x^4 + C_5x^5 + C_6x^6 + \ldots\ldots$

Put $x = 1, i, -1, -i$

$$2^n = C_0 + C_1 + C_2 + C_3 + C_4 + C_5 + C_6 + \ldots\ldots \qquad \ldots(1)$$

$$(1+i)^n = C_0 + C_1i - C_2 - C_3i + C_4 + C_5i - C_6 - C_7i + \ldots\ldots \qquad \ldots(2)$$

$$0^n = C_0 - C_1 + C_2 - C_3 + C_4 - C_5 + C_6 - C_7 \ldots\ldots\ldots\ldots \qquad \ldots(3)$$

$$(1-i)^n = C_0 - C_1i - C_2 + C_3i + C_4 - C_5i - C_6 + C_7i \ldots\ldots\ldots\ldots \qquad \ldots(4)$$

(1) – (2) + (3) – (4)

$$2^n - (1+i)^n - (1-i)^n = 4(C_2 + C_6 + C_{10} + \ldots\ldots)$$

$$2^n - 2^{\frac{n}{2}}\left[e^{\frac{in\pi}{4}} + e^{-\frac{in\pi}{4}}\right] = 4(C_2 + C_6 + C_{10} + \ldots\ldots)$$

$$2^n - 2^{\frac{n}{2}} \cdot 2\cos\frac{n\pi}{4} = 4(C_2 + C_6 + C_{10} + \ldots\ldots)$$

$$C_2 + C_6 + C_{10} + \ldots\ldots = \frac{1}{4}\left[2^n - 2^{\frac{n}{2}} \cdot 2\cos\frac{n\pi}{4}\right]$$

Paragraph for Question Nos. 29 to 30

29. If $w(\neq 1)$ is a cube root of unity and $(1+w)^7 = A + Bw$, then A and B are respectively the numbers :

यदि $w(\neq 1)$ इकाई का घनमूल है तथा $(1+w)^7 = A + Bw$, तो A तथा B क्रमशः है :

(A) 0 , 1 (B) 1 , 1 (C) 1 , 0 (D) –1 , 1

Ans. B

Sol. $(1+w)^7 = A + Bw$

$1 + w = -w^2$

$-w^{14} = A + Bw$

$-w^2 = A + Bw$

$1 + w = A + Bw$

$A = 1, B = 1$

30. If $(w \neq 1)$ is a cube root of unity then $\begin{vmatrix} 1 & 1+i+w^2 & w^2 \\ 1-i & -1 & w^2-1 \\ -i & -i+w-1 & -1 \end{vmatrix} =$

यदि $(w \neq 1)$ इकाई का घनमूल है, तो $\begin{vmatrix} 1 & 1+i+w^2 & w^2 \\ 1-i & -1 & w^2-1 \\ -i & -i+w-1 & -1 \end{vmatrix} =$

(A) 0 (B) 1 (C) i (D) w

Ans. A

Sol. $\begin{vmatrix} 1 & 1+i+w^2 & w^2 \\ 1-i & -1 & w^2-1 \\ -i & -i+w-1 & -1 \end{vmatrix}$

$C_3 \to C_3 + C_1$

$\begin{vmatrix} 1 & i-w & w^2 \\ 1-i & -1 & w^2-1 \\ 1-i & -1 & w^2-1 \end{vmatrix} = 0$

MATCH THE COLUMN

31. Match the statements given in Column I with the values given in Column II : **[JEE-2011]**

Column I	**Column II**
(A) If $\vec{a} = \hat{j} + \sqrt{3}\hat{k}, \vec{b} = -\hat{j} + \sqrt{3}\hat{k}$ and $\vec{c} = 2\sqrt{3}\hat{k}$ form a triangle, then the internal angle of the triangle between $\vec{a}$ and $\vec{b}$ is :	(p) $\frac{\pi}{6}$
(B) If $\int_a^b (f(x) - 3x)\,dx = a^2 - b^2$, then the value of $f\left(\frac{\pi}{6}\right)$ is	(q) $\frac{2\pi}{3}$
(C) The value of $\frac{\pi^2}{\ln 3}\int_{7/6}^{5/6} \sec(\pi x)\,dx$ is	(r) $\frac{\pi}{3}$
(D) The maximum value of $\left\lvert Arg\left(\frac{1}{1-z}\right)\right\rvert$ for $\lvert z\rvert = 1, z \neq 1$ is given by	(s) π
	(t) $\frac{\pi}{2}$

Column I में दिये गये कथनों को Column II में दिये गये मानों के साथ सूमेलित कीजिये : **[JEE-2011]**

Column I	**Column II**
(A) यदि $\vec{a} = \hat{j} + \sqrt{3}\hat{k}, \vec{b} = -\hat{j} + \sqrt{3}\hat{k}$ तथा $\vec{c} = 2\sqrt{3}\hat{k}$ एक त्रिभुज बनाते है, तो $\vec{a}$ तथा $\vec{b}$ के मध्य आन्तरिक कोण होंगा :	(p) $\frac{\pi}{6}$
(B) यदि $\int_a^b (f(x) - 3x)\,dx = a^2 - b^2$, है तो $f\left(\frac{\pi}{6}\right)$ का मान होगा	(q) $\frac{2\pi}{3}$
(C) $\frac{\pi^2}{\ln 3}\int_{7/6}^{5/6} \sec(\pi x)\,dx$ का मान होगा	(r) $\frac{\pi}{3}$
(D) $\left\lvert Arg\left(\frac{1}{1-z}\right)\right\rvert$, जहां $\lvert z\rvert = 1, z \neq 1$, का अधिकतम मान है	(s) π
	(t) $\frac{\pi}{2}$

Ans. (A) q (B) p or pqrs and t; (C) s (D) t

Sol. (A) $\vec{a} - \vec{b} = -1 + 3 = 2$

$|\vec{a}| = 2, |\vec{b}| = 2$

$\cos\theta = \frac{2}{2\times 2} = \frac{1}{2}$

$Q = \frac{\pi}{3}, \frac{2\pi}{3}$ but its $\frac{2\pi}{3}$ as its opposite to side of maximum length.

(B) $\int_a^b \left(f(x) - 3x\right) dx = a^2 - b^2$

$$\int_a^b f(x)\,dx = \frac{3}{2}\left(b^2 - a^2\right) + a^2 - b^2 = \frac{-a^2 + b^2}{2}$$

$\Rightarrow f(x) = x$

(C) $$\frac{\pi^2}{\ell n 3}\left(\frac{\ell n\left|\sec \pi x + \tan \pi x\right|_{7/6}^{5/6}}{\pi}\right) = \frac{\pi}{\ell n 3}\left(\ell n\left|\sec\frac{5\pi}{6} + \tan\frac{5\pi}{6}\right| - \ell n\left|\sec\frac{7\pi}{6} + \tan\frac{7\pi}{6}\right|\right) = \pi$$

(D) Let $u = \frac{1}{1-z} \Rightarrow z = 1 - \frac{1}{u}$

$$|z| = 1 \Rightarrow \left|1 - \frac{1}{u}\right| = 1$$

$\Rightarrow |u - 1| = |u|$

$\Rightarrow$ locus of u is perpendicular bisector of line segment joining 0 and 1.

$\Rightarrow$ maximum arg u approaches $\frac{\pi}{2}$ but will not attain.

32. Match the statements given in Column I with the intervals/union of intervals given in column II : **[JEE-2011]**

Column I	**Column II**
(A) The set $\left\{\text{Re}\left(\frac{2iz}{1-z^2}\right); z \text{ is a complex}, \lvert z\rvert = 1, z \neq \pm 1\right\}$ is	(p) $(-\infty, -1) \cup (1, \infty)$
(B) The domain of the function $f(x) = \sin^{-1}\left(\frac{8(3)^{x-2}}{1-3^{2(x-1)}}\right)$ is	(q) $(-\infty, 0) \cup (0, \infty)$
(C) If $f(\theta) = \begin{vmatrix} 1 & \tan\theta & 1 \\ -\tan\theta & 1 & \tan\theta \\ -1 & -\tan\theta & 1 \end{vmatrix}$ then the set $\left\{f(\theta) : 0 \le \theta < \frac{\pi}{2}\right\}$ is	(r) $[2, \infty)$
(D) If $f(x) = x^{3/2}(3x - 10), x \ge 0$ then f(x) is increasing in	(s) $(-\infty, -1] \cup [1, \infty)$
	(t) $(-\infty, 0] \cup [2, \infty)$

Column I में दिये गये कथनों को Column II में दिये गये अन्तरालों के साथ सूमेलित कीजिये : **[JEE-2011]**

Column I	**Column II**
(A) समुच्चय $\left\{\text{Re}\left(\frac{2iz}{1-z^2}\right); z \text{ d I ffeJ I ka; k, } \lvert z\rvert=1, z\neq\pm1\right\}$ है	(p) $(-\infty,-1)\cup(1,\infty)$
(B) फलन $f(x)=\sin^{-1}\left(\frac{8(3)^{r-2}}{1-3^{2(x-1)}}\right)$ का प्रान्त होगा	(q) $(-\infty,0)\cup(0,\infty)$
(C) यदि $f(\theta)=\begin{vmatrix}1 & \tan\theta & 1\\ -\tan\theta & 1 & \tan\theta\\ -1 & -\tan\theta & 1\end{vmatrix}$ है तो समुच्चय $\left\{f(\theta): 0\le\theta<\frac{\pi}{2}\right\}$ है	(r) $[2,\infty)$
(D) यदि $f(x)=x^{3/2}(3x-10), x\ge0$ है तो f(x) वर्धमान होगा	(s) $(-\infty,-1]\cup[1,\infty)$
	(t) $(-\infty,0]\cup[2,\infty)$

Ans. $A\to s; B\to t; C\to r; D\to r$

Sol. (A) $z=\dfrac{2i(x+iy)}{1-(x+iy)^2}=\dfrac{2i(x+iy)}{1-(x^2-y^2+2ixy)}$

Using $1-x^2=y^2$

$$Z=\frac{2ix-2y}{2y^2-2ixy}=-\frac{1}{y}$$

$$\because -1\le y\le1\Rightarrow -\frac{1}{y}\le-1 \text{ or } -\frac{1}{y}\ge1$$

(B) For domain

$$-1\le\frac{8\cdot3^{x-2}}{1-3^{2(x-1)}}\le1$$

$$\Rightarrow -1\le\frac{3^x-3^{x-2}}{1-3^{2x-2}}\le1$$

Case–I : $\dfrac{3^x-3^{x-2}}{1-3^{2x-2}}-1\le0$

$$\Rightarrow\frac{(3^x-1)(3^{x-2}-1)}{(3^{2x-2}-1)}\ge0$$

$\Rightarrow x \in (-\infty, 0] \cup (1, \infty)$

Case–II : $\dfrac{3^x - 3^{x-2}}{1 - 3^{2x} - 2} + 1 \geq 0$

$\Rightarrow \dfrac{\left(3^{x-2} - 1\right)\left(3^x + 1\right)}{\left(3^x \cdot 3^{x-2} - 1\right)} \geq 0$

$\Rightarrow x \in (-\infty, 1) \cup [2, \infty)$

So, $x \in (-\infty, 0] \cup [2, \infty)$

(C) $R_1 \to R_1 + R_3$

$$f(\theta) = \begin{vmatrix} 0 & 0 & 2 \\ -\tan\theta & 1 & \tan\theta \\ -1 & -\tan\theta & 1 \end{vmatrix}$$

$= 2\left(\tan^2\theta + 1\right) = 2\sec^2\theta \qquad \Rightarrow 2\sec^2\theta \in [2, \infty)$

(D) $f'(x) = \dfrac{3}{2}(x)^{\frac{1}{2}}(3x - 10) + (x)^{\frac{3}{2}} \times 3 = \dfrac{15}{2}(x)^{\frac{1}{2}}(x - 2)$

Increasing, when $x \geq 2$